HCR demo

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Overview

During ACLIM phase 2 (2019-2022), modelers evaluated a suite of Harvest Control Scenarios (1-5), in 2025 in collaboration with GOA-CLIM2 we added a number of additional HCRs to the set. Below is a list of those standardized harvest control rules and the equations used to derive the curves.

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ABC+HCR 1: Status quo

This is the basic sloping harvest control rule for groundfish in the EBS. There is a B20% cut-off for SSL (Atka, pollock, P. cod). $F_{ABC_{max}}$ is the HCR adjusted F rate that corresponds to ABC. The Tier three approach is to set the slope of the sloping HCR to $\alpha = 0.05$ and $B_{lim} = 0$ and $B_{target} = B_{40\%}$ or $B_{target} = 0.4\ B_0$ (i.e., 40% of unfished biomass B_0, as an MSY proxy) for most species except $B_{lim} = 0.2B_0$ ($B_{20\%}$) for pollock and Pacific cod .

Eq. 1

$$F_{ABC_{max}} = \left[\begin{array}{ll} F_{ABC} & \frac{B_y}{B_{target}} > 1 \\ F_{ABC}((\frac{B_y}{B_{target}} - \alpha)/(1-\alpha)) & \frac{B_{lim}}{B_{target}} \leq \frac{B_y}{B_{target}} < 1 \\ 0 & \frac{B_y}{B_{target}} < \frac{B_{lim}}{B_{target}} \end{array} \right.$$

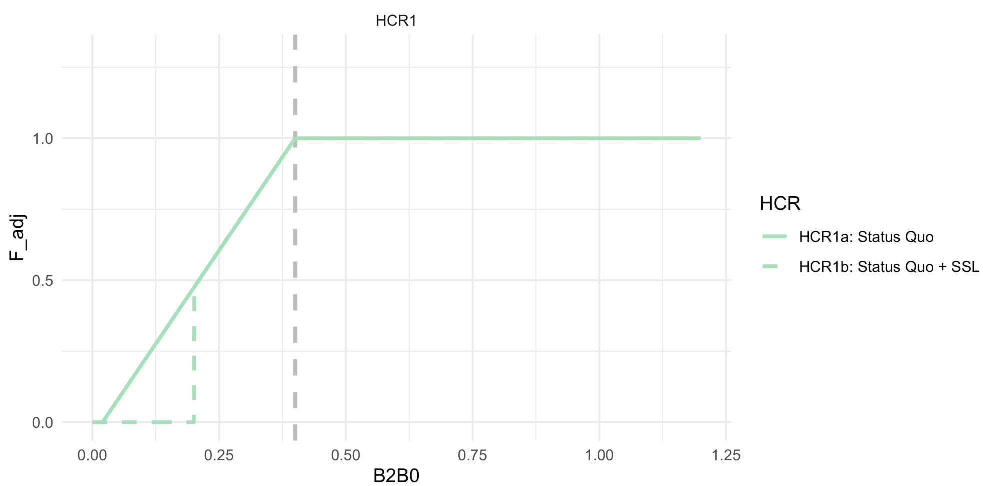


Figure 1: ABC+HCR 1: Status quo. This is the Tier 3 Harvest Control Rule, including the $B_{20\%}$ cuttoff for certain species

ABC+HCR 2: Lagged recovery to estimate emergency relief financing needs

This simulation set will help us estimate the approximate cost of emergency relief funds by artificially closing the fishery at $B_{25\%}$ (mimicking an economic-driven closure). During recovery to mimic lagged fishery recovery from a closure shock, we further delay harvest recovery by adjusting F rate using a larger alpha during the recovery period. Implementation of this would be use the difference in revenue between HCR2 and HCR1 to shorten the recovery period following a shock, e.g., by using the estimate to build a “rainy day” fund to supplement the fishery during climate shocks.

This is the same as in HCR 1 except that the fishery shuts down earlier at $B_{lim} = B_{25\%}$ and during the simulated lagged recovery the alpha is steeper (slower recovery; $\alpha = 0.30$ instead of $\alpha = 0.05$

Details: Sloping HCR with $B_{target} = B_{40\%}$ $\alpha = 0.05$, $B_{lim} = $,0.25 i.e., the cutoff to simulate where one might initiate emergency $ (i.e., simulate the fishery stopping operations at 0.25 due to CPUE and cost limitations, well before management closes the fishery based on SSB) and a steeper $ = $ 0.3 to simulate a lagged recovery (recovery occurs at $B_{40\%}$). Calculate difference in catch relative to HCR1 to get an estimate of what $ relief (“rainy day fund”) would be needed to supplement the fishery in closures.

Eq. 1

$$F_{ABC_{max}} = \begin{cases} F_{ABC} & \frac{B_y}{B_{target}} > 1 \\ F_{ABC}((\frac{B_y}{B_{target}} - \hat{\alpha}_y)/(1-\hat{\alpha}_y)) & \frac{B_{lim}}{B_{target}} \leq \frac{B_y}{B_{target}} < 1 \\ 0 & \frac{B_y}{B_{target}} < \frac{B_{lim}}{B_{target}} \end{cases}$$

and,

$$\hat{\alpha}_y = \begin{cases} \alpha \quad if \ \ \hat{\alpha}_{y-1} = \alpha \ \ \| \ \ \frac{B_y}{B_{target}} \geq 1 \\ \alpha_r \quad if \ \ \hat{\alpha}_{y-1} = \alpha_r \ \| \ \ \frac{B_y}{B_{target}} < \frac{B_{low}}{B_{target}} \end{cases}$$

For ACLIM HCR2 model runs we set α to 0.05 and α_r to 0.3, whith $ B_{low} = B_{lim} = 0.25B_{0}$.

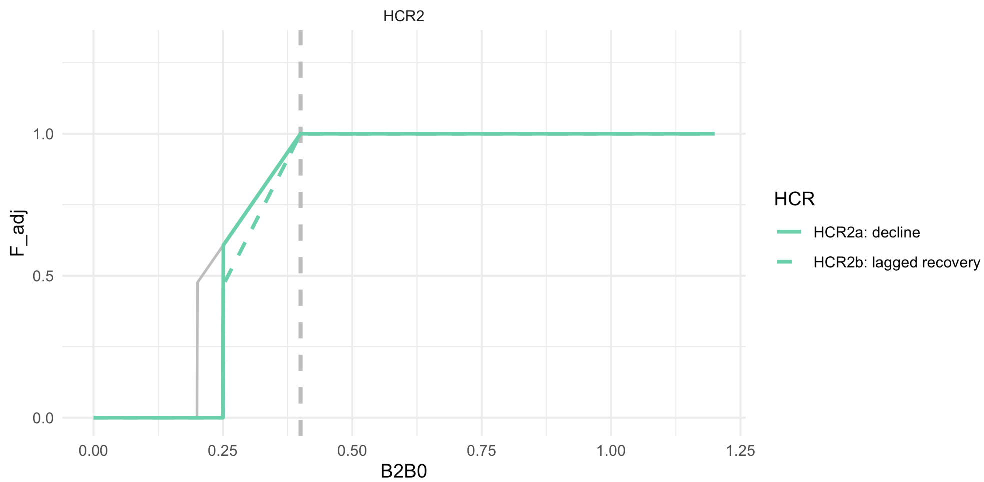


Figure 2: ABC+HCR 2: Lagged recovery to estimate emergency relief financing needs

ABC+HCR 3: Long-term resilience (stronger reserve) Ftarget

This is the same as in HCR 1 except that $B_{target} = B_{50\%}$.

Details: For HCR3a, set parameters to those as in HCR1a ($\alpha = 0.05$, $B_{lim} = 0$) except set B_{target} to 50% of B_0 instead of 40% B_0 . For HCR3b, use the same approach but set the B_{lim} to 20% of B_0 for amendment 80 species. Here, we are testing whether this would result in more stable biomass levels and catches over time and under alternative future conditions.

ABC+HCR 4: CE informed sloping rate, e.g., MHW category alpha

This is the same as in HCR 1 except that the proposed approach would scale back harvest rates faster below B_target for species that are climate sensitive, or during MHWs. Set the target to 40% of B_0 ($\alpha = 0.05$, $B_{lim} = B_{20\%}$, $B_{target} = B_{40\%}$) for normal conditions/climate resilient species. But for other species, below $B_{40\%}$ have steeper alphas based on MHW category forecasts for summer conditions (or alternatively, climate vulnerability ratings). This reduces future harvest intensity when conditions are forecast to be poor and could help rebound stocks faster following MHW. MHWs are characterized as Category 1-4 based on the

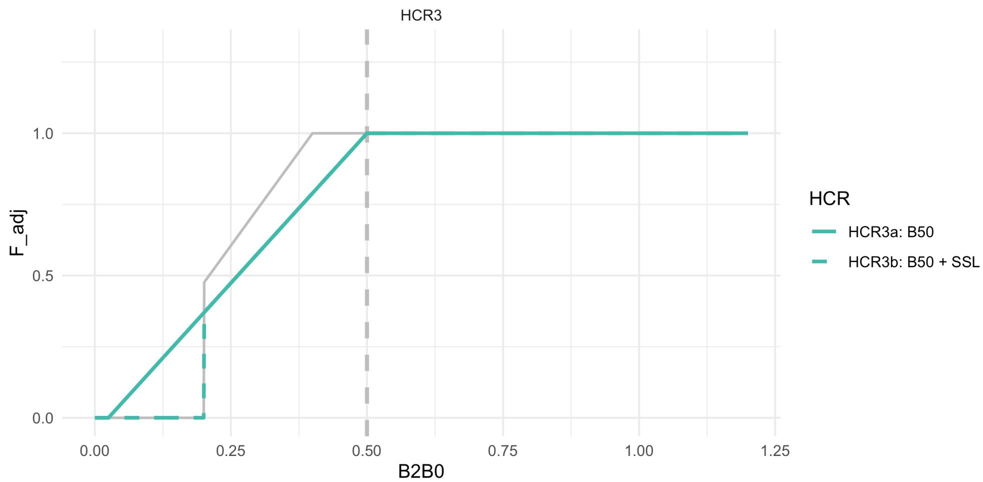


Figure 3: ABC+HCR 3: Long-term resilience (stronger reserve) Ftarget

degree of anomalous conditions above mean climatology (Category 1 = +1 standard deviations above the mean climatology, Category 4 = +4 SD).

Details: Set the target to 40% of B_0, ($\alpha = 0.05 + \text{MHW}_{category} * .09$, $B_{lim} = B_{20\%}$, $B_{target} = B_{40\%}$). E.g., shown, Category 2 MHW set the $\alpha = 0.1 * 2$ (for Category 2 MHW) = 0.2, for large MHW (category 4) set the $\alpha = 0.1 * 4 = 0.4$. We're testing whether this would result in more stable biomass levels and catches.

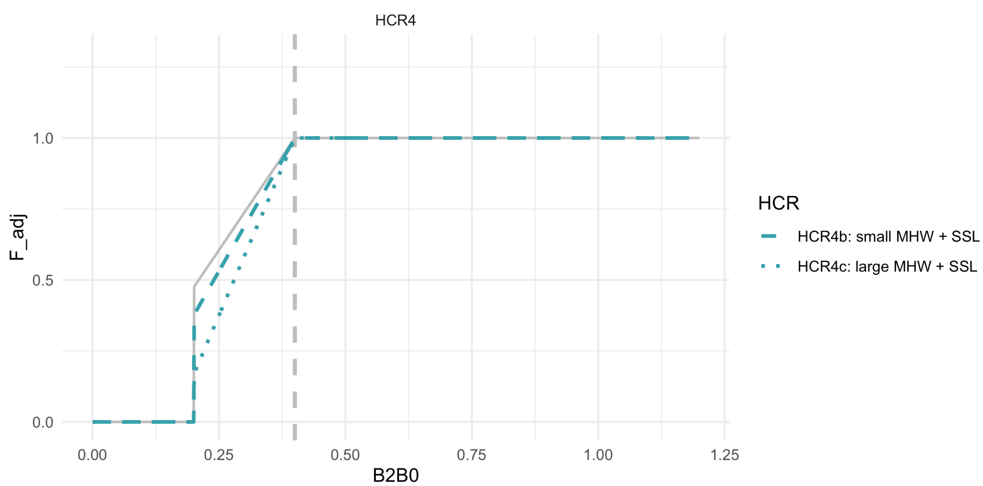
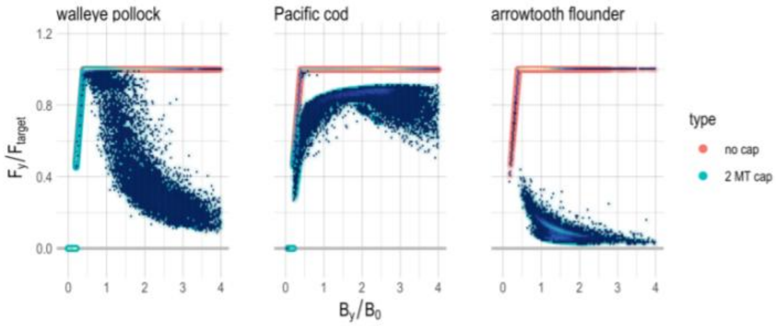


Figure 4: ABC+HCR 4: CE informed sloping rate, e.g., MHW category alpha

ABC+HCR 5: climate sensitivity reserve (buffer shocks)

The general idea here is to recreate the pollock cap effect when over B_{target}. The steepness of that cap effect could be varied based on vulnerability (or approximated via MSE), more sensitive species might need more reserve in the "bank". Pollock are an example of the HCR 5 in practice (via effects of the 2MT cap + sloping HCR).

Details: As in HCR1b, set the target to 40% of B_0 ($\alpha = 0.05$, $B_{lim} = B_{20\%}$, $B_{target} = B_{40\%}$).After $B_{40\%}$



Supplementary Figure 3. Effective harvest rate F_y under the no cap and 2 MT cap scenarios. Ratio of effective F_y to F_{target} as a function of the ratio of biomass to unfished biomass reference points (B_y/B_0) for each pollock (left), Pacific cod (middle), and arrowtooth flounder (right). Scenarios without the cap (orange; follow the ABC harvest control rule exactly) and scenarios with the 2 MT cap (scatter, teal).

Figure 5: Holsman et al. 2020 Figure

have a slowly sloping F proportional to climate sensitivity to mimic realized F rates of pollock under the 2 MT cap, i.e., reserve biomass for climate shocks sensu Holsman et al. 2020. In HCR 9 we use the same approach but use forecasts or MHW decadal predictions to set the right hand side of the curve above $B_{40\%}$.In this the climate sensitivity buffer γ is a value > 0 that controls the F decay rate (increase in reserve biomass) above B_{target}, i.e., $B_{40\%}$:

Eq. 1

$$F_{ABC_{max}} = \left[\begin{array}{ll} F_{ABC}\ e^{(-\gamma(\frac{B_y}{B_{target}}-1))} & \frac{B_y}{B_{target}} > 1 \\ F_{ABC}((\frac{B_y}{B_{target}} - \alpha)/(1-\alpha)) & \frac{B_{lim}}{B_{target}} \leq \frac{B_y}{B_{target}} < 1 \\ 0 & \frac{B_y}{B_{target}} < \frac{B_{lim}}{B_{target}} \end{array} \right.$$

where, $\quad 0 \leq \gamma \leq 1.$

Details: Set the biological reference points as in HCR1b, i.e., 40% of B_0, ($\alpha = 0.05$, $B_{lim} = B_{20\%}$, $B_{target} = B_{40\%}$). Above B_{target} apply the gamma / 2MT cap effects. Shown, for low sensitivity stocks set $\gamma = 0.1$, for highly sensitive stocks set $\gamma = 0.2$. We're testing whether this would result in more stable biomass levels and catches through shocks.

ABC+HCR 6: MHW slope + climate sensitivity reserve (buffer shocks)

The general idea here is to combine the HCR 4 MHW category 0-4 scaling factor when below B_target (B40) and a cap-like effect when over B_{target} (HCR5); values are as specified in those scenarios.

Details: Set the target to 40% of B_0, ($\alpha = 0.05$, $B_{lim} = B_{20\%}$, $B_{target} = B_{40\%}$). Shown, for low sensitivity stocks set $\gamma = 0.1$, for highly sensitive stocks set $\gamma = 0.2$. Shown as well, category 2 MHW set the $\alpha = 0.2$, for large MHW set the $\alpha = 0.4$.

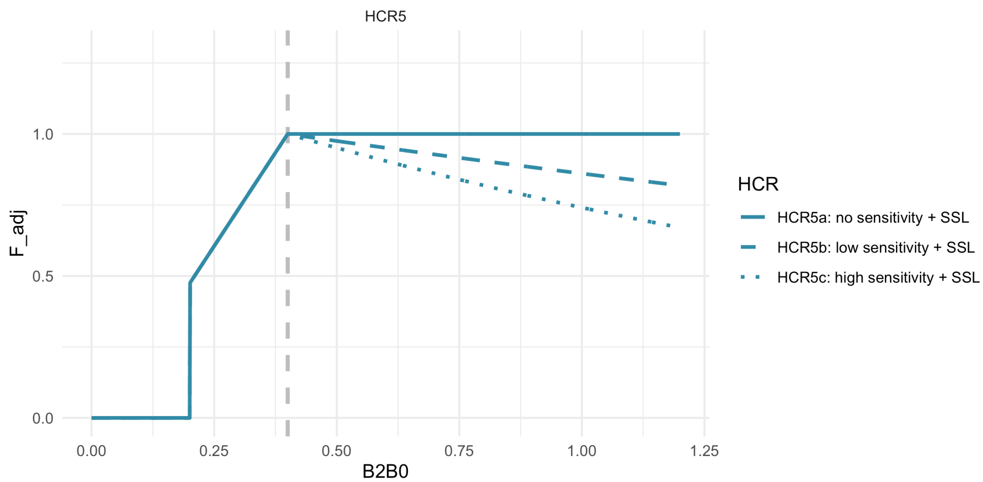


Figure 6: ABC+HCR 5: climate sensitivity reserve (buffer shocks)

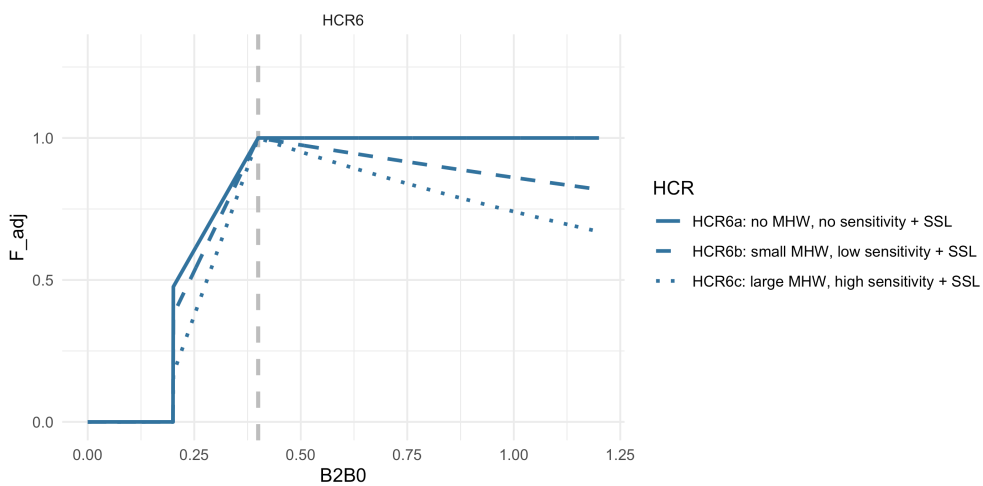


Figure 7: ABC+HCR 6: MHW slope + climate sensitivity reserve (buffer shocks)

ABC+HCR 7:Recruit per spawner biomass variability adjusted HCR based on covariate effects on S/R

This approach is based on recent analyses by P. Spencer (in prep) and formulated by K. Holsman. The general idea is to adjust the HCR ref points based on variability in SR relationships with a covariate X_y (e.g., SST) such that:

Eq. 1

$$F_{ABC_{max}} = \left[\begin{array}{ll} F_{ABC}\ e^{(\omega_1 * \mathrm{x_y})} & \frac{B_y}{\hat{B}_{target}} > 1 \\ F_{ABC}((\frac{B_y}{\hat{B}_{target}} - \alpha)/(1-\alpha))\ e^{(\omega_1 * \mathrm{x_y})} & \frac{B_y}{\hat{B}_{lim}} \leq \frac{B_y}{\hat{B}_{target}} < 1 \\ 0 & \frac{B_y}{\hat{B}_{target}} < \frac{\hat{B}_{lim}}{\hat{B}_{target}} \end{array} \right.$$

such that F_{ABC} is adjusted by covariate x_y according the parameter ω_1 and B_{target} and B_{lim} are adjusted based on the parameters ω_2 and ω_3 such that:

$$\hat{B}_{target} = B_{target} e^{(-\omega_2 * \mathrm{x_y})}$$

and

$$\hat{B}_{lim} = B_{lim} e^{(-\omega_3 * \mathrm{x_y})}$$

and ω_1, ω_2 and $\omega_3 \geq 0$. For the ACLIM HCR 7 simulations ω_1 and ω_2 will be fit using retrospecitve analyses of spawner recruitment relationships across sclaed (z-scored) SST (x_y) on EBS pollock by Spencer et al. in prep. Details: Set the target to 40% of B_0 ($\alpha = 0.05$, $B_{lim} = B_{20\%}$, $B_{target} = B_{40\%}$). In the example below (x_y) = 2.4 and for case 7a all ω values were set to 0.0, in case 7b $\omega_1 = \omega_2 = \omega_3 = 0.1$, and in case 7c ω_1 = -0.1, ω_2 = -0.1, ω_3 = -0.1.

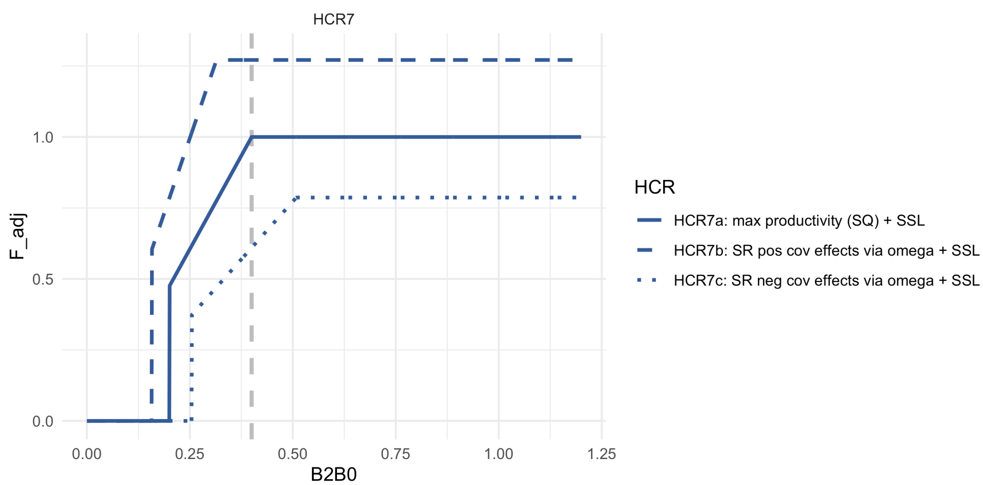


Figure 8: ABC+HCR 7:Recruit per spawner biomass variability adjusted HCR based on analyses by Spencer et al.

ABC+HCR 8: Adjust effective spawning biomass (rather than adjust B_target)

In this scenario, rather than adjust the FMP target from $B_{40\%}$, the effective B_y and B_{y+1} is adjusted downward:

Eq.

$$F_{ABC_{max}} = \left[\begin{array}{ll} F_{ABC} & \frac{\hat{B}_y}{B_{target}} > 1 \\ F_{ABC}((\frac{\hat{B}_y}{B_{target}} - \alpha)/(1-\alpha)) & \frac{B_{lim}}{B_{target}} \leq \frac{\hat{B}_y}{B_{target}} < 1 \\ 0 & \frac{\hat{B}_y}{B_{target}} < \frac{B_{lim}}{B_{target}} \end{array} \right.$$

where, $\hat{B}_y = \theta B_y$ and θ is a scaler on B_y (could be set to a value or could be a function of environmental covariates).

Details: Set the target to 40% of B_0 ($\alpha = 0.05$, $B_{lim} = B_{20\%}$, $B_{target} = B_{40\%}$). In the example below (x_y) = 2.4 and for case 8a all θ is set to 1, in case 8b $\theta = 0.8$.

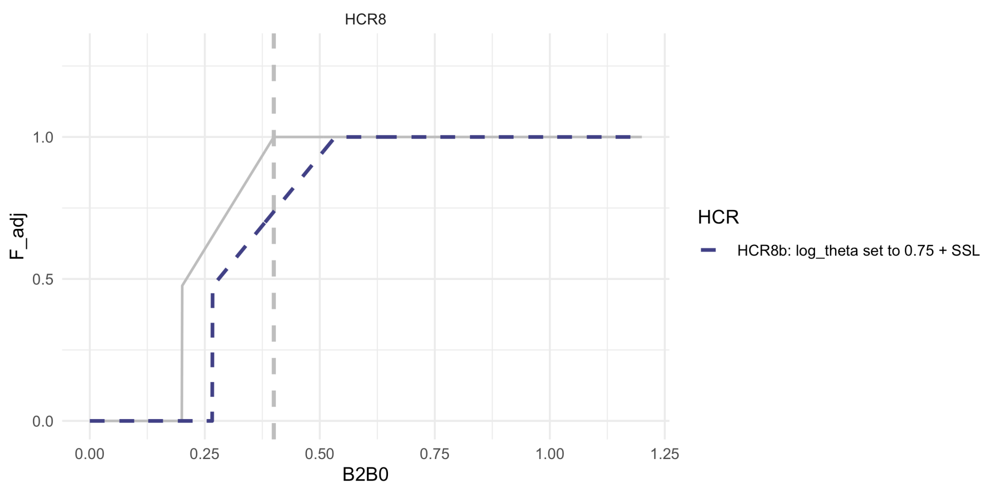


Figure 9: ABC+HCR 8:Adjust effective spawning biomass (rather than adjust B_target)

ABC+HCR 9: Forecast informed version of HCR 5

As in HCR 5 and 6 the approach here is to recreate the effect on pollock F rates when over B_{target}. The steepness of that cap effect here is tied to z-score scaled forecasted conditions X_y (e.g., next year SST, shown below as larger reserved needed when SST is warmer than average; or an average index for MHW categories forecasted for the next 5 years) , more sensitive species might need a stronger gamma to drive a larger age class reserve in the "bank". Pollock are an example of the HCR 5 in practice (via effects of the 2MT cap + sloping HCR).

Eq. 1

$$F_{ABC_{max}} = \left[\begin{array}{ll} F_{ABC}\ e^{(-\gamma(\text{inv.logit}(-X_y)/.5)(\frac{B_y}{B_{target}}-1))} & \frac{B_y}{B_{target}} > 1 \\ F_{ABC}((\frac{B_y}{B_{target}} - \alpha)/(1-\alpha)) & \frac{B_{lim}}{B_{target}} \leq \frac{B_y}{B_{target}} < 1 \\ 0 & \frac{B_y}{B_{target}} < \frac{B_{lim}}{B_{target}} \end{array} \right.$$

where, $0 \leq \gamma \leq 1$.

Details: As in HCR1b, set the target to 40% of B_0, ($\alpha = 0.05$, $B_{lim} = B_{20\%}$ for amendment80 species, $B_{target} = B_{40\%}$). Shown, set $\gamma = 3.7$. We're testing whether this would result in more stable biomass levels and catches through upcoming shocks. In the example below (x_y) = 2.4

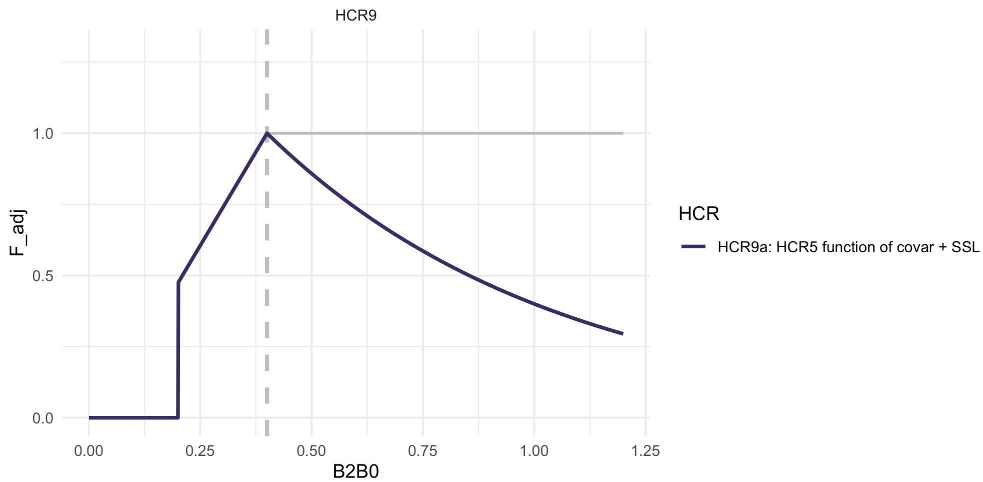


Figure 10: ABC+HCR 9:Forecast informed version of HCR 5

ABC+HCR 10: Forecast informed version of HCR 5

As in HCR 5, 6, and 9 the approach here is to recreate the effect on pollock F rates when over B_{target}. The steepness of that cap effect here is inverse to SSB, with an optional B_target offset (point at which F reduction starts), more sensitive species might need a lower gamma to drive a larger age class reserve in the "bank" sooner. Pollock are an example of the HCR 10 in practice (via effects of the 2MT cap + sloping HCR).

Eq. 1

$$F_{ABC_{max}} = \left[\begin{array}{ll} F_{ABC}/(\frac{B_y}{B_{target}}\frac{1}{(1+\gamma)}) & \frac{B_y}{B_{target}} > (1+\gamma) \\ F_{ABC} & 1 < \frac{B_y}{B_{target}} < (1+\gamma) \\ F_{ABC}((\frac{B_y}{B_{target}} - \alpha)/(1-\alpha)) & \frac{B_{lim}}{B_{target}} \leq \frac{B_y}{B_{target}} < 1 \\ 0 & \frac{B_y}{B_{target}} < \frac{B_{lim}}{B_{target}} \end{array} \right.$$

where, $\quad 0 \leq \gamma \leq 1$.

Details: As in HCR1b, set the target to 40% of B_0, ($\alpha = 0.05$, $B_{lim} = B_{20\%}$ for amendment80 species, $B_{target} = B_{40\%}$). Shown, no offset ("a") where $\gamma = 0$, small offset ("b") $\gamma = 0.2$, and ("c") large offset $\gamma = 0.5$. We're testing whether this would result in more stable biomass levels and catches through upcoming shocks. In the example below $(x_y) = 2.4$

Full set of HCR scenarios

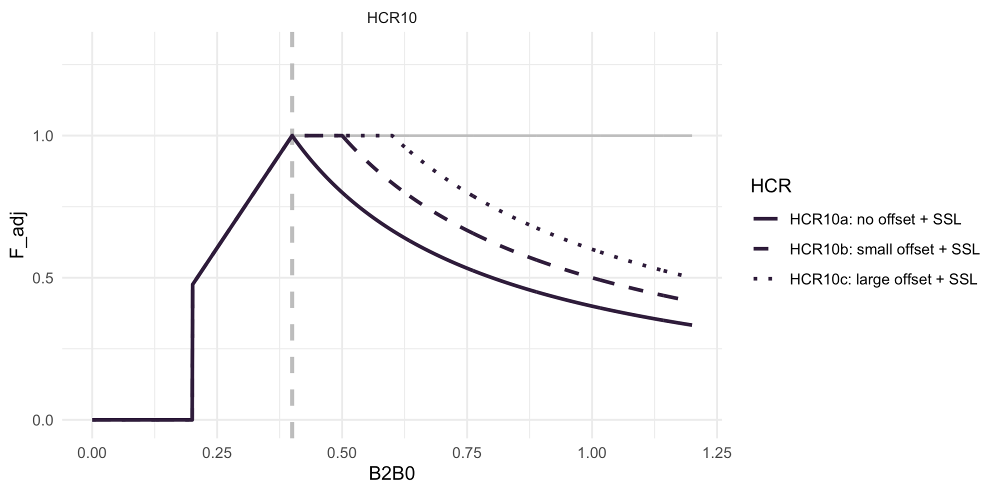


Figure 11: ABC+HCR 10: gamma offset on inverse SSB decay rate

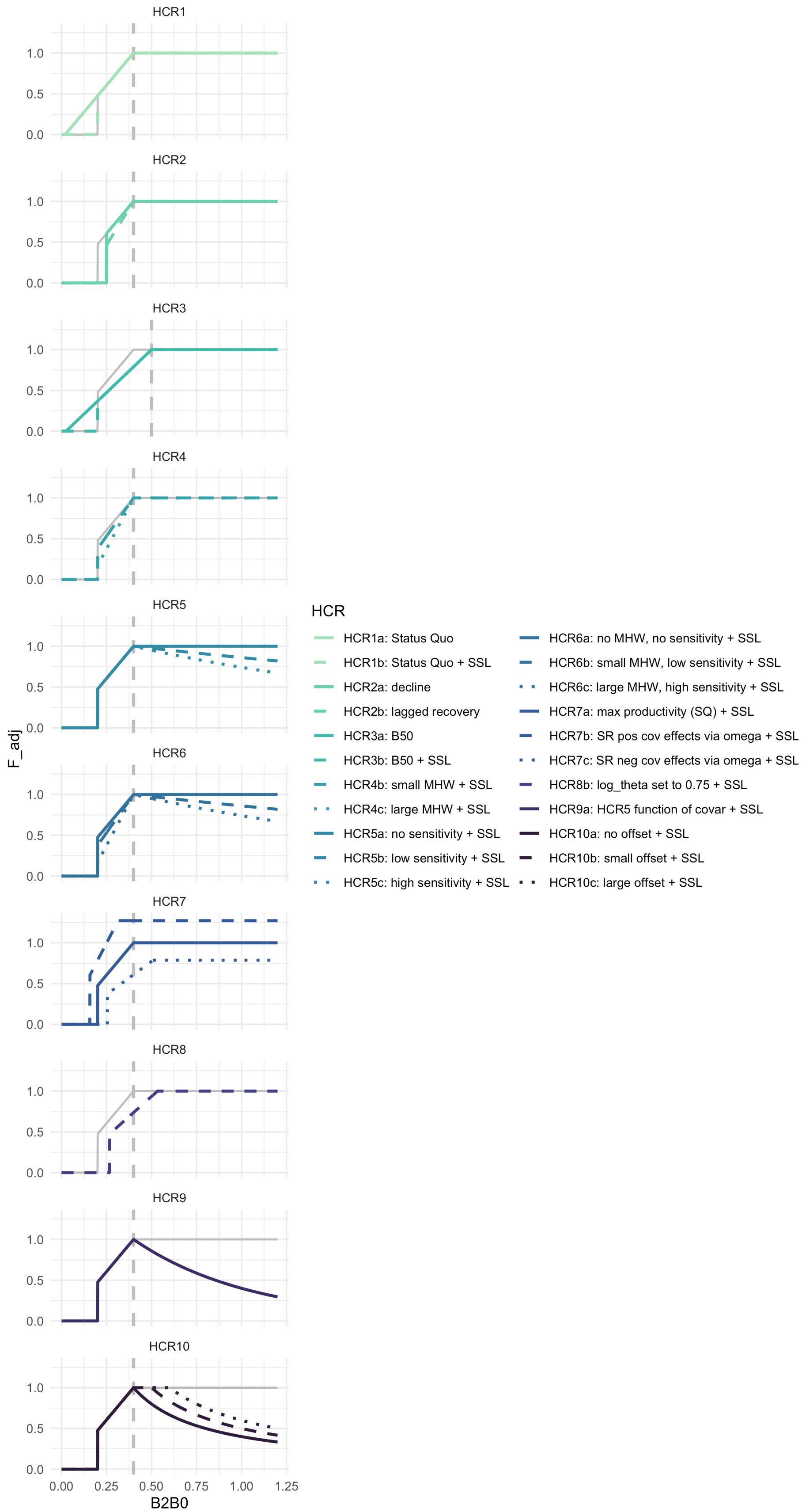


Figure 12: ABC+HCR 1- 9: Full set of HCR scenarios